

Deepakumar V U



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Areas of Expertise

- **Languages:** Golang, Python
- **Cloud:** AWS (Lambda, API Gateway, SQS/SNS/SES, IoT Core, CloudWatch, WAF)
- **Databases:** DynamoDB, Timestream, PostgreSQL, ClickHouse, FalkorDB
- **Security & Systems:** PKI, AWS Private CA, RBAC, Distributed Systems, IoT
- **DevOps & Platform:** Docker, Kubernetes, Jenkins, CloudFormation, AWS SAM
- **AI-Assisted Development:** Claude Code, Cursor

Work Experience

Senior Software Engineer

Tuskira Inc.

August 2025 – Present

Pune, Maharashtra

Building an AI-powered security platform integrating 150+ tools — focused on authorization, threat intelligence pipelines, and scalable data systems across a multi-repo Go/Python architecture.

Tech Stack: Golang, Python, AWS, PostgreSQL, ClickHouse, Temporal, Kubernetes, Kafka, gRPC/Protobuf

- Optimized **data-intensive dashboards (GB-scale)** using ClickHouse, reducing API latency by **70%**.
- Designed and implemented a **full RBAC authorization system** with **SSO** via AWS Cognito (SAML/OIDC), **JWT-based** stateless policy evaluation with **condition-based filtering**, enforced via gateway middleware across all platform services.
- Improved the zero-day **threat intelligence pipeline**, optimized **AI agent prompt efficiency**, fixed finding deduplication, and debugged production alert enrichment across the full controller-to-ClickHouse data flow.
- Designed notification and alerting systems processing **millions of alerts** with automated JIRA/ServiceNow ticket creation, including structured description rendering per vendor format.
- Built asynchronous export and report generation systems using **Temporal workflows**, handling **millions of records** with S3 presigned URL delivery and zip-compressed CSV output.
- Introduced a **generic normalization layer** for data ingestion, extracting vendor-specific asset fields into a unified tagging schema across **150+ integration sources**.

Senior Cloud Backend Engineer

Espressif Systems

January 2024 – July 2025

Pune, Maharashtra

Owned and scaled the Matter DAC Service (managed PKI) for IoT devices at scale.

Tech Stack: Golang, AWS, Jenkins, Python, GitLab

- Owned and scaled the **Matter DAC Service (managed PKI)** supporting **millions of IoT devices**.
- Reduced PKI infrastructure cost by **90%** by designing a custom solution over AWS KMS.
- Designed and maintained **distributed backend systems** for high-scale IoT workloads.
- Built a **flexible RBAC system** across products, improving access control and security.
- Drove **engineering quality and best practices**, including high test coverage and code standards.
- Mentored engineers and drove **code quality and engineering best practices**.

Cloud Backend Engineer

Espressif Systems

July 2021 – December 2023

Pune, Maharashtra

Led design and delivery of a scalable cloud-based PKI system in 8 weeks, earning the ESP Champion Club Award (Q3 2023).

Tech Stack: Golang, AWS, Jenkins, Python, GitLab

- Implemented various optimizations like **dependency management** to **eradicate** production build **errors**, dependency **vulnerability screening** and setup Gitlab pipeline for merge requests.
- Designed and delivered a **scalable PKI system** end-to-end in **8 weeks**, earning the **ESP Champion Club Award (Q3 2023)**.
- Built **serverless, event-driven architectures** on AWS for IoT features (group sharing, stats, time-series systems).
- Improved performance (**20% faster**) and reduced costs (**30%**) via batching and concurrency optimizations.
- Developed **time-series and data processing systems** using DynamoDB and Timestream.
- Established **scenario-based testing (>95% coverage)** and improved CI/CD reliability through dependency management and vulnerability scanning.

Associate Cloud Engineer
Espressif Systems

December 2020 – June 2021
Pune, Maharashtra

Developed backend features for RainMaker and resolved a critical security vulnerability.

Tech Stack: Golang, AWS

- Developed backend features for **RainMaker IoT platform** using Golang.
- Identified and fixed a **critical privilege escalation vulnerability**, strengthening system security.
- Improved developer productivity by introducing tools and workflows to streamline development.

Software Engineer
UBS

July 2019 – November 2020
Pune, Maharashtra

Streamlined risk simulation models using R and reduced debugging time by 80% with a custom error-logging package. Integrated DevOps with TeamCity and Puppet, cutting deployment time by 50% and issues by 90%.

Tech Stack: R, Oracle MySQL DB, Python, Teamcity, Puppet, Shell Scripting

Education

Pune Institute of Computer Technology
B.E. Computer Engineering

August 2015 – June 2019
Pune, Maharashtra

- **CGPA: 8.88/10.0**
- Coding and Financial lead of PICT Robotics 2018.
- **Project:** Automated Glaucoma detection using CNN, sponsored by National Institute of Ophthalmology (NIO).

Positions

Vice Chair
Connectivity Standards Alliance

October 2023 – July 2025
Pune, Maharashtra

- Acting as a Vice Chair of Security Design Tiger Team.
- Leading DA-PKI Revocation for Matter.

Certifications

AWS Certified Solutions Architect – Professional
Issued June 2025 - Expires June 2028

AWS Certified Solutions Architect – Associate
Issued June 2022 - Expires June 2028